

Trainings ▾

Remove

⚠ WARNING ⚠

To reduce the possibly of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

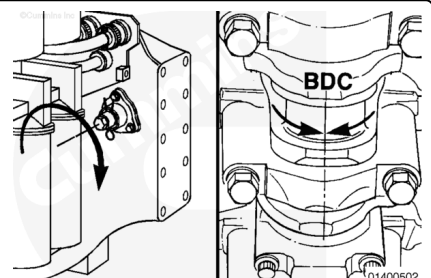
Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Drain the oil and remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).
- Remove the oil suction tube and block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1 (</qs3/pubsys2/xml/en/procedures/89/89-001-046.html>).



Remove

Use the barring mechanism to rotate the engine. Rotate the crankshaft to position a connecting rod at bottom dead center (BDC).

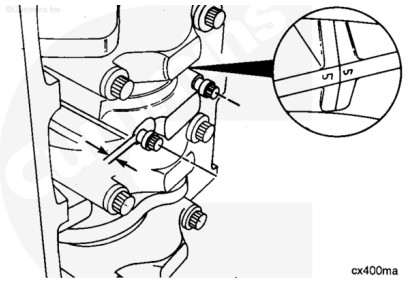


⚠ CAUTION ⚠

**Remove the piston cooling nozzles prior to piston removal.
Damage to the piston nozzles can result in engine failure.**

The connecting rods **must** have the cylinder number marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for correct markings. Use a steel stamp to mark any rod that is **not** correctly marked.

Loosen the capscrews until there is a 6 mm [$\frac{1}{4}$ in] clearance between the rod cap and capscrew head.

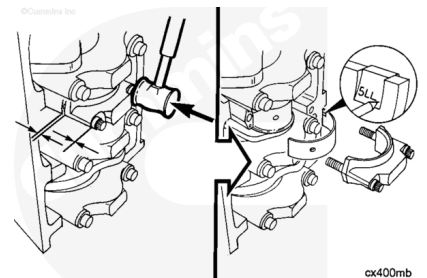


Use a mallet to tap the connecting rod capscrews until the connecting rod cap and rod separate.

Remove the capscrews and the cap.

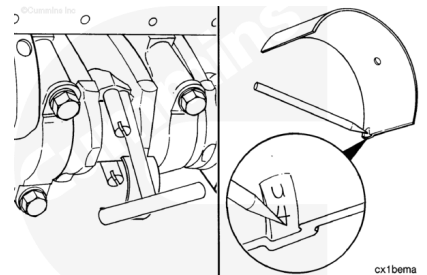
Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

Mark the cylinder number and the letter "L" in the flat surface of the bearing tangs.



Use a T-handle piston pusher to push the rod away from the crankshaft. Push the rod up far enough to allow the upper bearing shell to be removed.

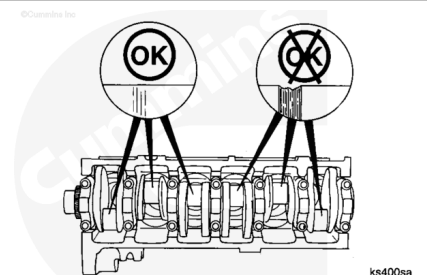
Remove the bearing shell and mark the cylinder number and the letter "U" in the flat surface of the bearing tangs.



Install

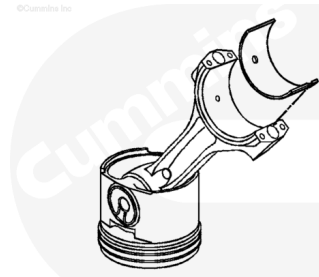
Use a clean, lint-free paper towel to clean the rod bore, the cap, and the crankshaft rod journal.

Inspect the crankshaft rod journals for damage.



⚠ CAUTION ⚠

Do not lubricate the back of the bearing shells. Engine damage can occur.



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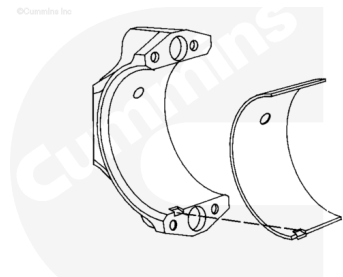
Note : The illustration shows the piston and connecting rod removed from the block for better clarity.

Used bearings **must** be installed in the same location from which they were removed.

Lubricate the bearing shell with clean engine oil. Install the upper bearing shell in the connecting rod with the tang of the bearing in the slot of the rod.

Pull the connecting rod down against the crankshaft journal.

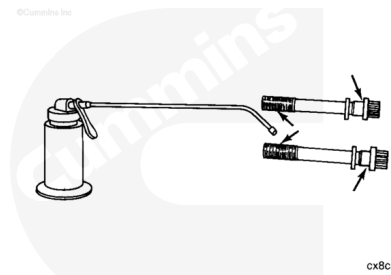
Install the lower bearing shell in the rod cap. Be sure the tang of the bearing shell is in the slot of the rod cap and the end of the bearing is even with the surface of the rod cap.



cx8behd

Use clean 15W-40 engine oil to lubricate the connecting rod capscrews and washers, as shown.

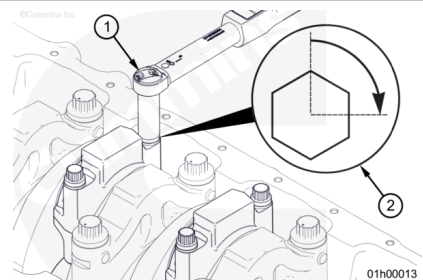
Install the washers and capscrews in the rod caps.



cx8csha

⚠ CAUTION ⚠

The connecting rods and caps are not interchangeable. The rods and caps are machined as an assembly. Failure will result if the parts are mixed.



01h00013

⚠ CAUTION ⚠

The cylinder number on the rod and cap must be the same. Failure will result if the parts are mixed.

⚠ CAUTION ⚠

The side of the cap with the cylinder number marking (bearing tang side) must be toward the camshaft. Failure will result if the parts are not installed correctly.

Install the connecting rod cap.

Always use the torque-turn method to tighten the connecting rod capscrews.

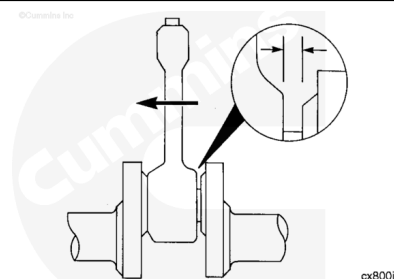
Use the following steps to tighten the capscrews alternately and evenly.

Torque Value:

1. 196 n•m [145 ft-lb]
2. Tighten 90°

Check the side clearance between the rod and the crankshaft.

Connecting Rod Side Clearance		
mm		in
0.20	MIN	0.008
0.37	MAX	0.015



The rod **must** move freely from side-to-side.

Finishing Steps

⚠ CAUTION ⚠

Do not pry or bend the suction tube to get the support bracket capscrew holes to align. Improper oil flow can cause engine damage.



- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1
(/qs3/pubsys2/xml/en/procedures/89/89-001-046.html).
- Install the oil suction tube and the block stiffener plate. Refer to Procedure 001-089 in Section 1
(/qs3/pubsys2/xml/en/procedures/89/89-001-089.html).
- Install the oil pan and related components. Refer to Procedure 007-025 in Section 7
(/qs3/pubsys2/xml/en/procedures/89/89-007-025.html).

- Use clean, 15W-40 engine oil to fill the engine. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>) for engine oil capacity.
- Operate the engine to 70°C [160°F] coolant temperature and check for leaks.

Last Modified: 11-Feb-2016
